Krithik Maran

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EDUCATION

The Ohio State University, College of Engineering — **Columbus, OH**
Bachelor of Science in Electrical and Computer Engineering — *May 2025*

- Specialization: Computer Engineering
- GPA: 3.51
- Relevant Coursework: Software Development and Data Structures, Systems, Discrete Structures and Runtimes, Digital Logic, Analog Systems, and Circuits, Microcontroller-Based Systems, Digital Time Signals, Computer Architecture

SKILLS

- Java, JavaScript, C/C++, Python, React.JS, MATLAB, HTML/CSS, MSP430 Assembly, SQL, Postgres, Pytorch, Scikit-learn, VHDL, Eclipse, VSCode, Code Composer Studio, Ignition SCADA

INDUSTRY EXPERIENCE

Honda Motor Company — **Columbus, OH**
Capstone Project Team Lead — *August 2024 – Present*

- Working on system integration of Matrox imaging library with Ignition SCADA.
- Creating a custom UI for Honda employees to interact with the Vision AI system.
- Creating a PostgreSQL database for archival of the last 2000 images processed.

Elevance Health — **Mason, OH**
DevOps Intern — *May 2024 – August 2024*

- Evaluated and built a proof of concept for integrating Argo Workflows, a container-native workflow engine for Kubernetes, to optimize DevOps teams' tasks.
- Scheduled and automated 4 service health-checking tasks using Argo Workflows to reduce DevOps strain by cutting down management and validation from hours to minutes.
- Designed and implemented an automated system using Argo Workflows to suspend and resume services in an AWS EKS cluster to facilitate infrastructure maintenance in the production environment.
- Assessed chaos engineering practices and tools and developed a blueprint to implement them in production to increase the reliability of services for members.
- Consolidated 15 different EKS, MSK, and ElasticCashe metrics into one easily accessible dashboard for the team using Datadog.

Ohio State University — **Columbus, OH**
Undergraduate Researcher — *May 2023 – May 2024*

- **Research through The Ohio State University Department of Welding Engineering for Los Alamos National Laboratory**
- Engineered a neural network using PyTorch, and Scikit-learn that predicts laser weld depths and widths based on given power and laser travel speed parameters.
- Optimized network with Optuna, a hyperparameter tuning framework, and integrated one-hot encoding to add functionality for different material types.
- Applied advanced preprocessing techniques, including cross-validation and scaling with Pandas and Scikit-learn, to ensure robust and reliable predictions from available data.
- Developed an ensemble learning algorithm to more accurately predict laser weld depths and widths by isolating weld nodes.

PROJECTS & INVOLVEMENT

TEK8 (Translating Engineering Research K-8) — **Columbus, OH**
Instructor — *August 2023 – December 2023*

- Taught engineering principles to underprivileged students at Metro Middle School.
- Created design challenges based on my laser welding research to help the students understand engineering concepts.

Destination Imagination Global Finals — **Kansas City, MO and Knoxville, TN**
Global Finals Contestant — *May 2018 — May 2019*

- Built a maze-navigating rover with a Raspberry Pi, Ultrasonic Sensor, and camera, and created a program to control it using SSH.
- Made a flying drone from scratch and configured the ESC setup, batteries, servo controls, and flight controller.